Brady Golomb

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EDUCATION

UNIVERSITY OF PENNSYLVANIA, School of Engineering and Applied Science — Philadelphia, PA
Master of Science in Engineering, Robotics (Accelerated Program) — Expected May 2028
Bachelor of Science in Engineering, Mechanical Engineering — Expected May 2028
Concentration: Dynamics, Controls, and Robotics | Minor: Mathematics — **Cumulative GPA: 3.84/4.0**

Relevant Coursework: Mechanical Design, Solid Mechanics, Thermal-Fluids Engineering, Controls, Advanced Dynamics, Feedback Control Design and Analysis, Design of Mechatronic Systems

EXPERIENCE

PENN ASSISTIVE DEVICES AND PROSTHETIC TECHNOLOGIES (ADAPT) — Philadelphia, PA — September 2024 – Present
Assistive Technology Engineer

- 2024 — Lead Engineer: diagnosed a recurring screw-stripping failure in a prosthetic bike-pedal adapter and redesigned it in SolidWorks; the adapter is now used successfully by cyclists, saving $1,000+/year in repairs
- Iterated through carbon-fiber prototypes and load-tested machined parts until the stripping failure was eliminated
- 2025–26 — Project Lead: leading a team with the Philadelphia Adaptive Sports Center to design an adaptive lacing system so athletes with physical disabilities can dress independently; prototypes in self-testing

ONE HEALTH PARTNERS — Naperville, IL — June 2026 – August 2026
Product Development Intern

- Developed Delta Wound App in Swift, an iOS app using the iPhone LiDAR sensor and AI image analysis to measure wounds and classify wound-bed tissue; now in TestFlight beta with clinicians
- Independently built an EMR discharge system that turns physician and clinician notes into plain-language patient audio briefings, adding no documentation burden for providers
- Shadowed 5 physicians across 5 specialties (50+ hours) to turn clinical workflow into product requirements

CARROLL ROBOTICS — Southlake, TX — August 2022 – January 2024
Robotics Team Lead and Lead Engineer

- Led design, fabrication, and programming of VEX V5 robots; placed 11th of 11,000+ teams globally in the Skills Competition and qualified for two World Championships
- Built a flywheel shooter and roller system modeled in Onshape; autonomous routines hit 94% reliability

PENN TRACK AND FIELD — Philadelphia, PA — August 2024 – Present
Walk-On Varsity Athlete

- Division I Ivy League athlete; scored at the Ivy League Championships (4th and 10th place)
- Train 20+ hours/week with a dual-degree course load; Wharton Leadership Academy and Student-Athlete Advisory Committee

PROJECTS

Running Form Robot — Personal Project — 2024 – Present

- Built an autonomous camera rover from a salvaged RC car that follows a runner outdoors; integrated AI motion-capture software to report stride rate in real time
- In progress: rebuilding as a ROS 2 (C++) and Gazebo gait-analysis pipeline, with a PID-controlled leg model and OpenCV pose tracking for stride rate and joint angles

Bottle Rocket Trajectory Model — MEAM 2480 — Spring 2026

- Built a Python flight model (thrust and ballistic phases) and calibrated its drag, nozzle, and butane coefficients to 28 test launches
- Wrote an inverse solver (grid search + Nelder-Mead) to choose launch angle and butane charge; the rocket cleared the target gap on its first attempt and landed within 2 m of the target

Rubber-Band Bungee Cord — MEAM 3470 — Fall 2026

- Characterized rubber bands in 18 MTS tensile tests and fit a cubic force model, validated on held-out tests
- Wrote a Python design model that searched band layouts against floor-clearance and g-limit constraints; predicted the demo-day fall within 2.7%

TECHNICAL SKILLS

CAD & Fabrication: SolidWorks, SolidWorks Simulation (FEA), Onshape, DraftSight, 3D Printing (incl. carbon fiber), Laser Cutting, Machining, Rapid Prototyping
Programming & Tools: Python (NumPy, SciPy), C++, MATLAB, Simulink, ROS 2, Linux, Arduino, Git, Java, OCaml, HTML/CSS
Awards & Certifications: Princeton Prize in Race Relations, Winner (2024); MathWorks MATLAB Onramp, Simulink Onramp (2026)